

II-Probabilistic Approaches

Course Summary

Foundations of Neuro-Symbolic AI

Alex Goessmann

University of Applied Science Würzburg-Schweinfurt (THWS)

July 6, 2026

Outline

- ▶ AI in the tensor formalism
- ▶ Neuro-Symbolic AI
- ▶ Logical Approaches
- ▶ Probabilistic Approaches

Definition of tensors

Definition (Tensor)

A tensor of order $d \in \mathbb{N}$ with leg dimensions $m_k \in \mathbb{N}$ (where $k \in [d]$) is a map

$$\tau [X_{[d]}] : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}.$$

The coordinate at the index tuple $x_{[d]}$ is the value $\tau [X_{[d]} = x_{[d]}]$.

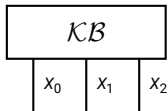
- ▶ The order of the tensor is the number d
- ▶ The number of coordinates is $\prod_{k \in [d]} m_k$

Visualization

Coordinatewise visualization:

$$\mathcal{KB}[X_0, X_1, X_2] = \begin{array}{c} \begin{array}{c} \xrightarrow{x_1} \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \downarrow x_0 \\ \begin{array}{c} 0 \\ 1 \end{array} \end{array} \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \begin{array}{c} \xrightarrow{x_2} \\ \begin{array}{c} 0 \\ 1 \end{array} \end{array} \end{array} [X_0, X_1, X_2]$$

Digrammatic visualization:

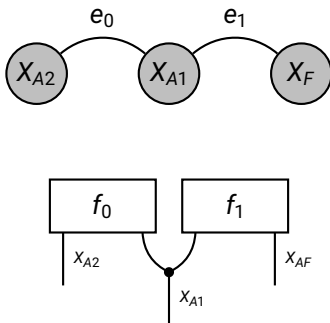


Tensor networks

We used the diagrammatic visualization to depict tensor networks on a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

- ▶ A set of nodes v , decorated by variables X_v
- ▶ A set of hyperedges $e \subset \mathcal{V}$ decorated by tensors $\tau^e [X_e]$

The toy accounting example was visualized by a hypergraph and a decoration by tensors as:



Definition of contractions

Definition (Contraction)

Let $\tau^{\mathcal{G}}$ be a tensor network on a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. For any subset $\mathcal{U} \subset \mathcal{V}$ we define the contraction to be the tensor

$$\langle \tau^{\mathcal{G}} \rangle_{[\mathcal{X}_{\mathcal{U}}]}$$

defined coordinatewise by the sum

$$\langle \tau^{\mathcal{G}} \rangle_{[\mathcal{X}_{\mathcal{U}} = \mathbf{x}_{\mathcal{U}}]} = \sum_{\mathbf{x}_{\mathcal{V} \setminus \mathcal{U}} \in \mathcal{X}_{\mathcal{V} \setminus \mathcal{U}}[m_{\mathcal{V}}]} \prod_{\mathbf{e} \in \mathcal{E}} \tau^{\mathbf{e}}[\mathbf{X}_{\mathbf{e}} = \mathbf{x}_{\mathbf{e}}] .$$

Tensor networks as a unifying language

Sparsity principle: Tensor network decompositions

- ▶ Logics: Syntactical description
- ▶ Neural: Circuit decompositions
- ▶ Probability: Representaion of independencies

Inference principle: Tensor network contractions

- ▶ Logics: Entailment
- ▶ Neural: Function evaluation
- ▶ Probability: Marginal and conditional distributions

Outline

- ▶ AI in the tensor formalism
- ▶ **Neuro-Symbolic AI**
- ▶ Logical Approaches
- ▶ Probabilistic Approaches

Symbolic AI

The symbolic paradigm

Humans reason in abstractions: Symbols

Advantages:

- ▶ Human interaction with models
- ▶ Guaranteed model behaviour

Challenges:

- ▶ Efficient model training
- ▶ Intractability of inference

Neural AI

The neural paradigm

Interesting functions decompose into networks of easy functions.

Basis encodings

- Represent arbitrary functions $q : \times_{k \in [d]} [m_k] \rightarrow \times_{\ell \in [p]} [n_\ell]$ as tensors:

$$\beta^q [Y_{[p]}, X_{[d]}] = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \epsilon_{q(x_{[d]})} [Y_{[p]}] \otimes \epsilon_{x_{[d]}} [X_{[d]}]$$

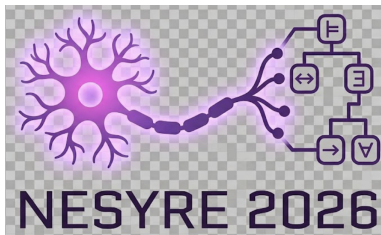
- Decomposition into circuits (see lecture I.6) is reflected in tensor network decompositions

Basis calculus

- Contractions with one-hot encodings infer the functions
- Local contractions along the direction of the circuit suffice

Come to our workshop!

My group in Berlin will organize a workshop in Berlin (September 14-17, 2026) on **Neuro-symbolic AI, Mathematical Reasoning and Agents**:



<https://www.wias-berlin.de/workshops/NESYRE2026/>

Please contact me if you would like to participate!
(alex.goessmann@wias-berlin.de)

Outline

- ▶ AI in the tensor formalism
- ▶ Neuro-Symbolic AI
- ▶ **Logical Approaches**
- ▶ Probabilistic Approaches

Syntactical decompositions

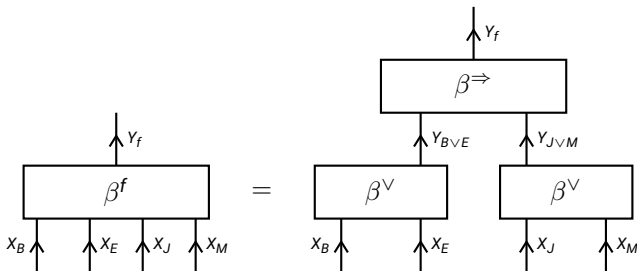
Sparsity by syntax

Syntactical representations of formulas provide means to decompose the tensor. The interesting formulas have syntactical representations of small size.

Example: The basis encoding of the formula

$$f[X_B, X_E, X_J, X_M] = (X_B \vee X_E) \Rightarrow (X_J \vee X_M)$$

is decomposed as



Logical Inference: Entailment

Definition (Entailment)

Let $\mathcal{KB} [X_{[d]}]$ and $f [X_{[d]}]$ be two propositional formulas. If for all worlds $x_{[d]}$, for which $\mathcal{KB} [X_{[d]} = x_{[d]}] = 1$, we also have $f [X_{[d]} = x_{[d]}] = 1$, then we say that f is entailed by $\mathcal{KB}$ and denote

$$\mathcal{KB} \models f.$$

Contraction Criterion: We have that a formula $\mathcal{KB} [X_{[d]}]$ entails another formula $f [X_{[d]}]$, if

$$\langle \mathcal{KB} [X_{[d]}], \neg f [X_{[d]}] \rangle_{[\emptyset]} = 0.$$

Outline

- ▶ AI in the tensor formalism
- ▶ Neuro-Symbolic AI
- ▶ Logical Approaches
- ▶ Probabilistic Approaches

Independence

Definition (Conditional Independence)

Given a joint distribution of variables X_A , X_B and X_C , we say that X_A and X_B are independent given X_C if for any index tuple (x_A, x_B, x_C)

$$\mathbb{P}[X_A = x_A, X_B = x_B | X_C = x_C] = \mathbb{P}[X_A = x_A | X_C = x_C] \cdot \mathbb{P}[X_B = x_B | X_C = x_C],$$

We denote this situation by

$$(X_A \perp X_B) | X_C.$$

Contraction criterion: We have $(X_A \perp X_B) | X_C$ if and only if

$$\mathbb{P}[X_A, X_B | X_C] = \langle \mathbb{P}[X_A | X_C], \mathbb{P}[X_B | X_C] \rangle_{[X_A, X_B, X_C]}.$$

Graphical models: Visualizing independencies in hypergraphs

Bayesian networks (directed hypergraphs):

- ▶ We choose any enumeration $[d]$ of the nodes respecting the direction (outgoing nodes after incoming nodes)
- ▶ To each directed hyperedge we define the incoming nodes as parents to the outgoing node
- ▶ Any Bayesian network on the hypergraph satisfies for each $k \in [d]$ the conditional independencies

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

Markov networks (undirected hypergraphs):

- ▶ Disjoint node sets A and B are separated by C , if any path from A to B passes a node in C .
- ▶ Any Markov network on the hypergraph satisfies for any node sets A and B separated by C

$$(X_A \perp X_B) \mid X_C$$